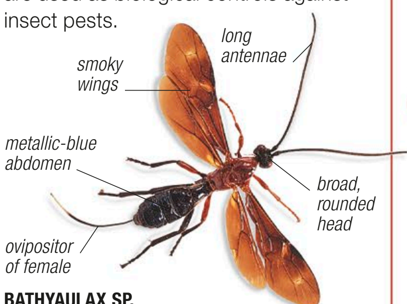
Order Hymenoptera *continued*

Family Braconidae

Braconid wasps

Occurrence 20,000 spp. worldwide; on suitable caterpillar hosts

Like ichneumon wasps (see below), braconids are parasitoids—animals that develop inside a living host, eventually killing it. Braconids mainly attack the caterpillars of moths and butterflies, often laying over 100 eggs on or in their victims. The developing larvae feed on the caterpillar's tissues, pupate, and fly away as adults. Adult braconids are typically less than $\frac{7}{32}$ in (7 mm) long, and reddish brown or black. Many species are used as biological controls against insect pests.



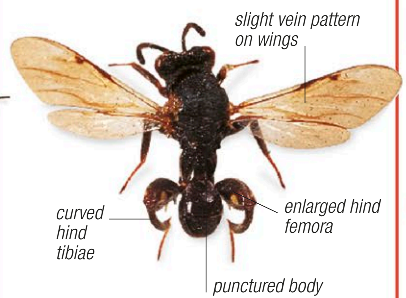
BATHYAULAX SP.
Larger than most of its family, these braconids measure $\frac{1}{2}$–$\frac{3}{4}$ in (1.5–2 cm), and are found in Africa and Southeast Asia. They are used to control stem-boring caterpillar pests of cereal crops.

Family Chalcididae

Chalcid wasps

Occurrence 1,800 spp. worldwide; on suitable insect hosts

Mostly under $\frac{5}{16}$ in (8 mm) long, chalcids make up a small but important family of parasitic insects. Many species lay their eggs in beetle larvae, or in the caterpillars of butterflies and moths, but some are hyperparasites, attacking the eggs or larvae of other parasitic insects. Adult chalcids are typically dark brown, black, red, or yellow, sometimes with a metallic sheen. Their larvae—like those of other wasps—are white and grublike, without legs. Some chalcid species are bred and released to control insect pests.



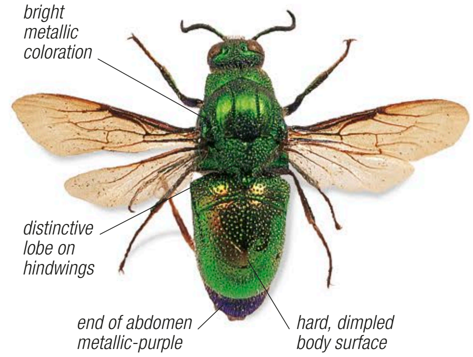
CHALCIS SISPES
This species is native to Europe and Asia. Its larvae parasitize the larvae of soldier flies.

Family Chrysididae

Jewel wasps

Occurrence 3,000 spp. worldwide; on suitable insect hosts

Jewel wasps, also known as cuckoo or ruby-tailed wasps, have very hard and dimpled bodies that protect them from bee and wasp stings. They are $\frac{1}{16}$–$\frac{3}{4}$ in (0.2–2 cm) long. Most species are a very bright metallic-blue, green, or red, or combinations of these colors. The abdomen is hollowed out underneath, allowing the wasp to curl up if attacked. Typically, the female lays an egg in the nest of a solitary bee or wasp, and the larva immediately hatches. The jewel wasp larva then feeds on the host larva and also eats the host's provisions. The adults feed on nectar and liquids.



STILBUM SPLENDIDUM
The splendid emerald wasp is a North Australian species that parasitizes solitary mud-nesting wasps.

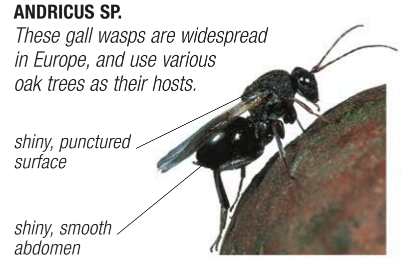
Family Cynipidae

Gall wasps

Occurrence 1,250 spp. mostly in the Northern Hemisphere; on trees, plants

Gall wasps are shiny, reddish brown or black, and usually fully winged. They measure $\frac{1}{32}$–$\frac{11}{32}$ in (1–9 mm) in length; males are usually smaller than females. The latter have a laterally flattened abdomen and a humped thorax. Many species, by a process not yet fully understood, induce the growth of species-specific galls on oaks and related trees by laying eggs inside the plant tissue. The galls protect and nourish the developing larvae. Life cycles of gall-forming species involve sexual and asexual generations.

ANDRICUS SP.
These gall wasps are widespread in Europe, and use various oak trees as their hosts.

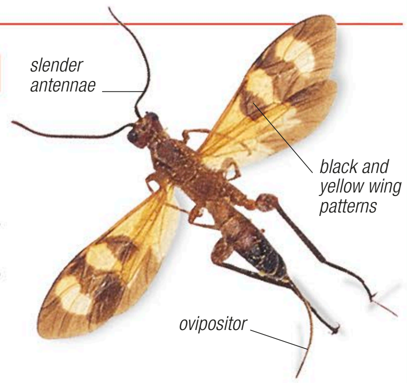


Family Ichneumonidae

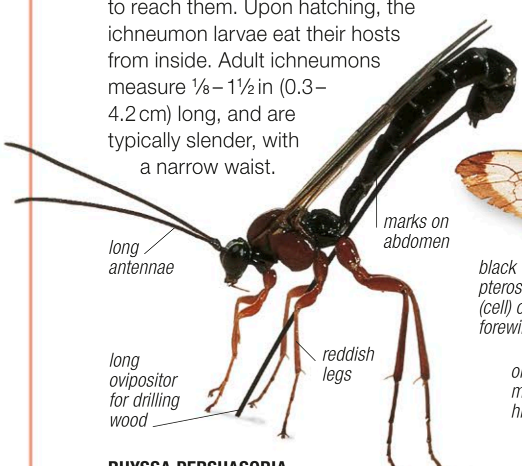
Ichneumon wasps

Occurrence 60,000 spp. worldwide, especially temperate regions; wherever suitable insect hosts are found

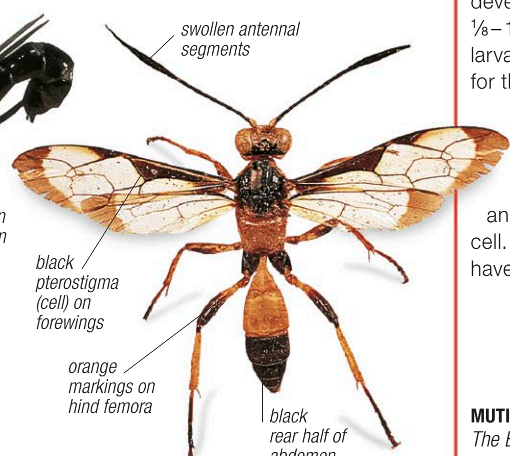
Ichneumon wasps are of great ecological importance, because they parasitize other insects, including many pests. Using their slender ovipositors, the females lay eggs in the larvae and pupae in a wide range of hosts, sometimes drilling through several inches of wood to reach them. Upon hatching, the ichneumon larvae eat their hosts from inside. Adult ichneumons measure $\frac{1}{8}$–$1\frac{1}{2}$ in (0.3–4.2 cm) long, and are typically slender, with a narrow waist.



PARACOLLYRIA SP.
These African wasps have a bright coloring that works as a warning to predators and is mimicked by other insects.



RHYSSA PERSUASORIA
The giant ichneumon wasp uses its long, drill-like ovipositor to reach the larvae of horntail sawflies deep in pine lumber.



JOPPA ANTENNATOR
This species is characterized by swollen antennal segments and contrasting, wasplike coloration.

Family Mutillidae

Velvet ants

Occurrence 4,000 spp. worldwide, especially subtropical and tropical regions; females found in dry habitats

Despite their name, these are not ants. The name velvet ant comes from the soft, velvety hairs that cover the antlike, wingless females; males have fully developed wings. Velvet ants are $\frac{1}{8}$–1 in (0.3–2.5 cm) long. They use the larvae of other wasps or bees as food for their larvae. Females bite open a suitable host cell and, if the larva in it is fully grown, lay their eggs on it and reseal the cell. The larvae then eat the host larva, and pupate inside the cell. Female velvet ants have powerful stings.

MUTILLA EUROPEA
The European velvet ant, widespread in the warmer regions of Europe, parasitizes species of bumble bee.

Family Mymaridae

Fairyflies

Occurrence 1,400 spp. worldwide; in a wide variety of habitats, wherever insect hosts are found

The smallest flying insects on earth belong to this family. Fairyflies are $\frac{1}{16}$–$\frac{3}{16}$ in (0.2–5 mm) long, and are dark brown, black, or yellow. The narrow front wings have a fringe of hairs but no obvious venation. The hindwings appear stalked, and are strap shaped. The females of all species parasitize the eggs of other insects, mostly of plant-hoppers and other bug families. Several fairyfly species are used as biological control agents against insect pests.

ANAGRUS OPTABILIS
This is a specialized parasitoid of the eggs of certain plant-hoppers. Related species have been used to control plant-hoppers that attack rice crops.